

Collections of various things

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1 Algebraic Geometry

1.1 2026-05-07

1.1:scheme is not affine

Prove that the open scheme obtained from $\text{Spec } k[x, y, z, w]/(xw - yz)$ removing codimension 1 subscheme $V(x, y)$ is not affine.

Let $X = \text{Spec } A = \text{Spec } k[x, y, z, w]/(xw - yz)$ and $U = X - V(x, y)$. We want to show that U is not affine. Note that A can be viewed as 2×2 matrix with determinant 0. Consider

$$\tilde{X} = \left\{ (x, y, z, w), (s : t) \mid \begin{pmatrix} x & z \\ y & w \end{pmatrix} \begin{pmatrix} s \\ t \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \subset \mathbb{A}^4 \times \mathbb{P}^1.$$

We have the projection to the first factor $\tilde{X} \rightarrow X$. Note that

$$\begin{pmatrix} x & z \\ y & w \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

is equivalent to $x = y = 0$. We define a map $\tilde{X} \dashrightarrow \mathbb{A}^1$ projecting onto $\{t \neq 0\}$, away from the point $(1 : 0) \in \mathbb{P}^1$, namely

$$\begin{aligned} \tilde{X} &\dashrightarrow \mathbb{A}^1, \\ (x, y, z, w), (s : t) &\mapsto \frac{s}{t} = -\frac{z}{x} = -\frac{w}{y}. \end{aligned}$$

This map factor through $U = X - V(x, y)$

$$\frac{z}{x} = \frac{w}{y} : X - V(x, y) = U \rightarrow \mathbb{A}^1$$

and we obtain an isomorphism given by

$$\begin{aligned} X - V(x, y) &\rightarrow (\mathbb{A}^2 - 0) \times \mathbb{A}^1, \\ (x, y, z, w) &\mapsto (x, y, \frac{z}{x} = \frac{w}{y}), \\ (x, y, xu, yu) &\leftarrow (x, y, u) \end{aligned}$$

Alternatively, we could compute $\mathcal{O}_X(U)$ via the covering $D(x) \cup D(y)$.

$$\begin{aligned}\mathcal{O}_X(D(x)) &= k[x, x^{-1}, y, z, w]/(xw - yz) = k[x, x^{-1}, y, z], \quad w = x^{-1}yz, \\ \mathcal{O}_X(D(y)) &= k[x, y^{-1}, y, z, w]/(xw - yz) = k[x, y^{-1}, y, w], \quad z = y^{-1}xw. \\ \mathcal{O}_X(D(xy)) &= k[x, y, x^{-1}, y^{-1}, z], \quad w = x^{-1}yz.\end{aligned}$$

And we know that

$$\mathcal{O}_X(U) = \ker(\mathcal{O}_X(D(x)) \oplus \mathcal{O}_X(D(y)) \rightarrow \mathcal{O}_X(D(xy)))$$

where the map is given by $(f, g) \mapsto f|_{D(xy)} - g|_{D(xy)}$ for $f \in \mathcal{O}_X(D(x))$ and $g \in \mathcal{O}_X(D(y))$. We claim that the element in the kernel $\mathcal{O}_X(U)$ must be of the form (a, b) such that $a \in k[x, y, \frac{z}{x}]$, $b \in k[x, y, \frac{w}{y}]$ and $a = b$ after mapping to the localization $\mathcal{O}_X(D(xy))$. This implies that $\mathcal{O}_X(U)$ is isomorphic to $k[x, y, \frac{z}{x} = \frac{w}{y}]$. If U is affine, then $U \simeq \mathbb{A}^3$. This cannot happen.

1.2: Cohomology and base change dimension jump up

The following is an example.